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Course: DSA0602-Data handling and visualization

Set 6: Product Sales Analysis

1. Create a grouped bar chart to visualize the sales of each product for the first quarter. Label the chart elements.

Program:

```
# Load libraries

library(ggplot2)

library(tidyr) #

Create dataset

sales <-
data.frame(
  Product_ID = c(1,2,3),
  Product_Name = c("Product A","Product B","Product C"),
  January = c(2000,1500,1200),
  February = c(2200,1800,1400),
  March = c(2400,1600,1100)
)

# Convert to long format

sales_long <- pivot_longer(
  sales, cols = c(January,
February, March), names_to =
"Month", values_to = "Sales"
)

# Grouped Bar Chart

ggplot(sales_long,
aes(x = Product_Name,
      y = Sales,      fill = Month)) +
geom_bar(stat = "identity", position =
```

```
"dodge") + labs( title = "First Quarter
Product Sales", x = "Product", y =
"Sales"

) +
theme_minimal()
```

Output:



2. Generate a stacked area chart to represent the overall sales trend for all products over the three months.

Program:

```
# Load libraries

library(ggplot2)

library(tidyr) #
Create dataset
sales <-
data.frame(

  Product_Name = c("Product A","Product B","Product C"),
  January = c(2000,1500,1200),
  February = c(2200,1800,1400),
  March = c(2400,1600,1100)
)

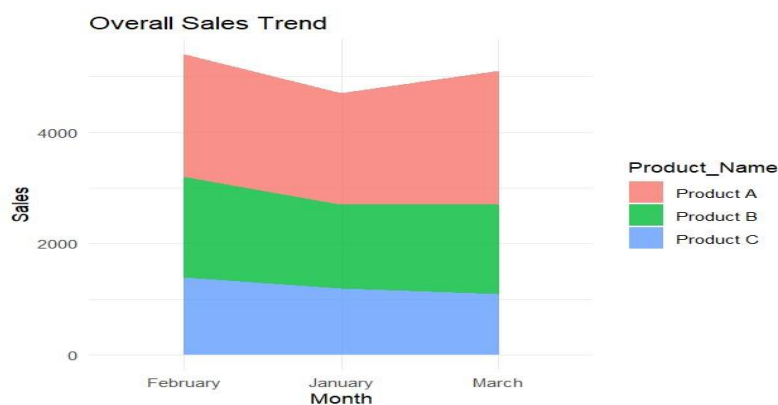
# Convert to long format
sales_long <- pivot_longer(
```

```

sales, cols = c(January,
February, March), names_to =
"Month", values_to = "Sales"
)
# Order months
sales_long$Month <- factor(
sales_long$Month, levels =
c("January","February","March
")
)
# Stacked Area Chart
ggplot(sales_long, aes(x =
Month, y = Sales,
fill = Product_Name,
group = Product_Name)) +
geom_area(alpha = 0.8) +
labs(title = "Overall
Sales Trend", x =
"Month", y = "Sales"
) +
theme_minimal()

```

Output:



3. Build a table to show the monthly sales data for each product. Label the table elements.

Program:

```
# Create dataset

sales <-

data.frame(

  Product_ID = c(1,2,3),

  Product_Name = c("Product A","Product B","Product C"),

  January_Sales = c(2000,1500,1200),

  February_Sales = c(2200,1800,1400),

  March_Sales = c(2400,1600,1100)

)

# Display table

print(sales)
```

Output:

```
Product_Name January February March
1 Product A    2000    2200  2400
2 Product B    1500    1800  1600
3 Product C    1200    1400  1100
```

Set 7: Customer Demographics Analysis

1. Create a bar chart to visualize the distribution of customer ages. Label the axes and title the chart.

Program:

```
# Load library

library(ggplot2) #

Create dataset

customers <-

data.frame(

  Customer_ID = c(1,2,3),

  Age = c(28,35,42),

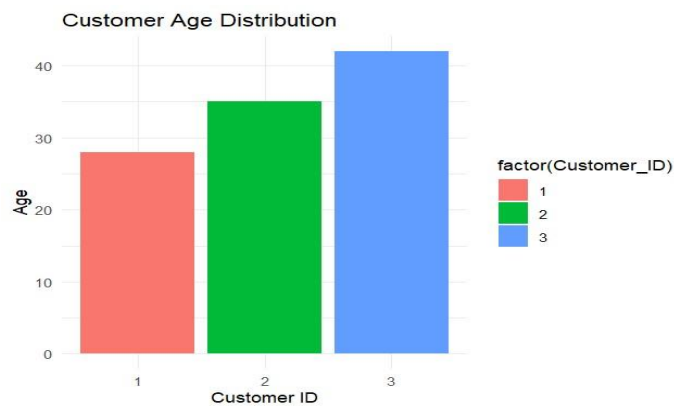
  Gender = c("Female","Male","Female"),

  Income = c(50000,60000,75000)

)
```

```
# Bar Chart ggplot(customers,
aes(x = factor(Customer_ID),
y = Age, fill =
factor(Customer_ID))) +
geom_bar(stat = "identity") +
labs(title = "Customer Age
Distribution", x = "Customer
ID", y = "Age"
) +
theme_minimal()
```

Output:



2. Generate a pie chart to represent the distribution of customers by gender.

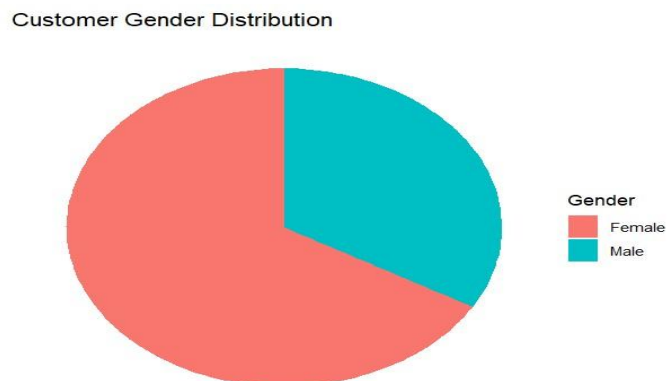
Program:

```
# Load libraries
library(ggplot2) #
Create dataset
customers <-
data.frame(
  Customer_ID = c(1,2,3),
  Gender = c("Female","Male","Female")
)
```

```
# Count genders gender_count <-
as.data.frame(table(customers$Gender))
colnames(gender_count) <- c("Gender","Count")

# Pie Chart
ggplot(gender_count,
       aes(x = "", y = Count, fill
= Gender)) + geom_bar(stat = "identity",
width = 1) + coord_polar("y") +
labs(title = "Customer Gender
Distribution") + theme_void()
```

Output:



3. Build a table to show the demographic information for each customer. Label the table elements.

Program:

```
# Create dataset
customers <-
data.frame(
  Customer_ID = c(1,2,3),
  Age = c(28,35,42),
  Gender = c("Female","Male","Female"),
  Income = c(50000,60000,75000)
)
```

```
# Display table  
print(customers)
```

Output:

```
Customer_ID Gender  
1      1 Female  
2      2  Male  
3      3 Female
```

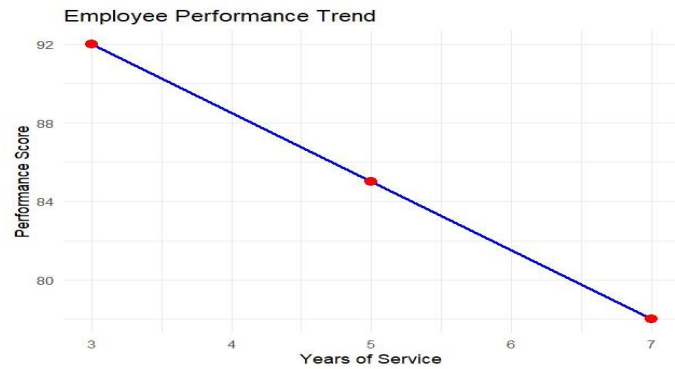
Set 8: Employee Performance Analysis

1. Create a line chart to visualize the performance trend of employees over time. Label the axes and title the chart.

Program:

```
# Load library  
library(ggplot2)  
# Create dataset  
employee <- data.frame(  
  Employee_ID = c(1, 2, 3),  
  Department = c("Sales", "HR", "Marketing"),  
  Years_of_Service = c(5, 3, 7),  
  Performance_Score = c(85, 92, 78)  
)  
# Line Chart ggplot(employee, aes(x = Years_of_Service, y =  
Performance_Score, group = 1)) + geom_line(color = "blue", linewidth =  
1) + geom_point(color = "red", size = 3) + labs( title = "Employee  
Performance Trend", x = "Years of Service", y = "Performance Score"  
) +  
theme_minimal()
```

Output:

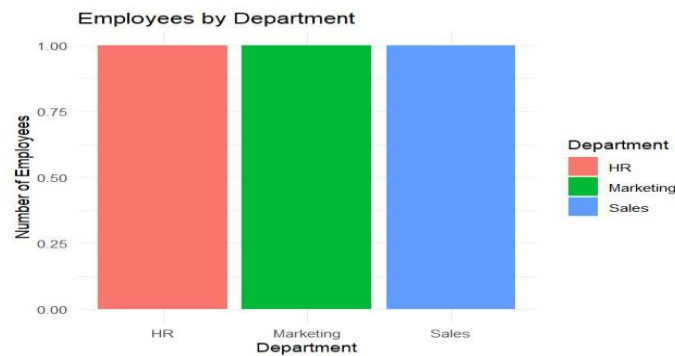


2. Generate a bar chart showing the distribution of employees across different departments. Label the chart elements.

Program:

```
# Load library
library(ggplot2) #
Create dataset
employee <-
data.frame(
  Employee_ID = c(1, 2, 3),
  Department = c("Sales", "HR", "Marketing"),
  Years_of_Service = c(5, 3, 7),
  Performance_Score = c(85, 92, 78)
)
# Bar Chart
ggplot(employee,
aes(x = Department,
fill = Department)) +
geom_bar() +
labs( title = "Employees by
Department", x = "Department",
y = "Number of Employees"
) +
theme_minimal()
```

Output:



3. Build a table to display the performance data for each employee. Label the table elements.

Program:

```
# Create dataset
```

```
employee <-
```

```
data.frame(
```

```
  Employee_ID = c(1, 2, 3),
```

```
  Department = c("Sales", "HR", "Marketing"),
```

```
  Years_of_Service = c(5, 3, 7),
```

```
  Performance_Score = c(85, 92, 78)
```

```
)
```

```
# Display table
```

```
print(employee) Output:
```

```
Employee_ID Department Years_of_Service Performance_Score
```

```
1      1    Sales          5           85
```

```
2      2     HR           3           92
```

```
3      3 Marketing        7           78
```

set 9: Online_Learning_Activity.csv

1. Import the dataset in R. Plot a histogram for Quiz_Score. Draw a boxplot of Quiz_Score by Course. Add title and labels. Interpret student performance.

Program:

```
# Load library
```

```
library(ggplot2) #
```

```
Create dataset
```

```

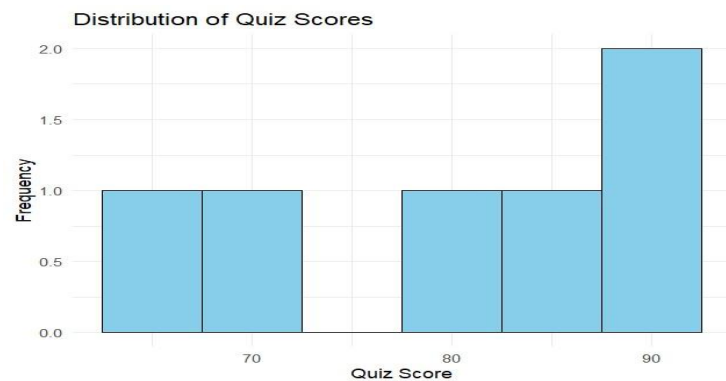
learning <-
data.frame(
  Student_ID = c("L01","L02","L03","L04","L05","L06"),
  Gender = c("Male","Female","Male","Female","Male","Female"),
  Age = c(20,22,19,21,23,20),
  Course = c("R","R","SQL","R","R","SQL"),
  Study_Time = c(3.5,4.2,2.0,5.0,2.5,4.0),
  Videos_Watched = c(12,15,8,18,9,14),
  Quiz_Score = c(78,85,65,92,70,88),
  Login_Date = c("2025-01-05","2025-01-05","2025-02-08",
    "2025-02-08","2025-03-12","2025-03-12")
)

# Histogram ggplot(learning, aes(x = Quiz_Score)) +
geom_histogram(binwidth = 5, fill = "skyblue", color =
"black") + labs(  title = "Distribution of Quiz Scores",  x
= "Quiz Score",  y = "Frequency"
) +
theme_minimal() #

Boxplot
ggplot(learning,
aes(x = Course,
y = Quiz_Score,
fill = Course)) +
  geom_boxplot() +
  labs(  title = "Quiz Scores
by Course",  x = "Course",
y = "Quiz Score"
) +
theme_minimal()

```

Output:



2. Create a scatter plot using R programming between Study_Time and Quiz_Score.

Represent

Videos_Watched using bubble size. Apply transparency. Explain the learning pattern.

Program:

```
# Load library
```

```
library(ggplot2) #
```

```
Create dataset
```

```
learning <-
```

```
data.frame(
```

```
  Student_ID = c("L01","L02","L03","L04","L05","L06"),
```

```
  Study_Time = c(3.5,4.2,2.0,5.0,2.5,4.0),
```

```
  Quiz_Score = c(78,85,65,92,70,88),
```

```
  Videos_Watched = c(12,15,8,18,9,14)
```

```
)
```

```
# Bubble Scatter Plot ggplot(learning,
```

```
  aes(x = Study_Time,      y =
```

```
    Quiz_Score,      size =
```

```
    Videos_Watched)) +
```

```
  geom_point(alpha = 0.6, color =
```

```
    "blue") + labs(  title = "Study Time
```

```
vs Quiz Score",  x = "Study Time
```

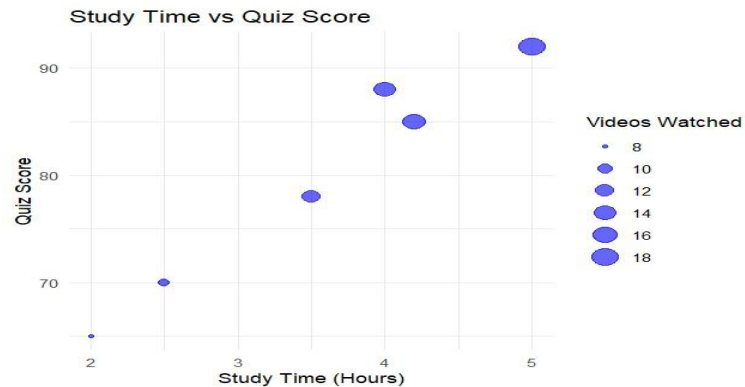
```
(Hours)",  y = "Quiz Score",  size =
```

```
    "Videos Watched"
```

```
) +
```

```
theme_minimal()
```

Output:



3. Using R programming, convert Login_Date into Date format. Calculate average quiz score per month. Plot a line chart. Apply moving average smoothing. Identify the trend.

Program:

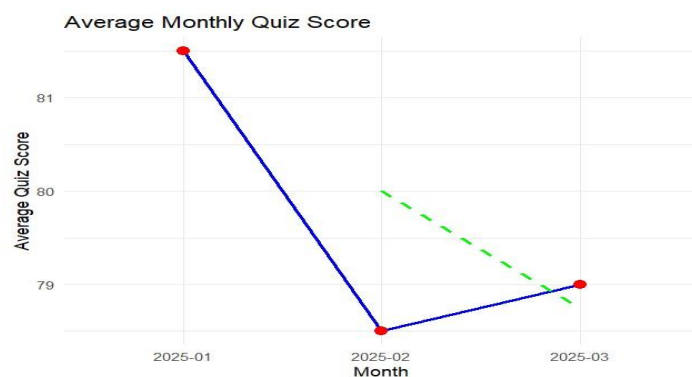
```
# Load libraries
library(ggplot2)
library(dplyr) #
Create dataset
learning <-
data.frame(
  Student_ID = c("L01","L02","L03","L04","L05","L06"),
  Quiz_Score = c(78,85,65,92,70,88),
  Login_Date = c("2025-01-05","2025-01-05","2025-02-08",
    "2025-02-08","2025-03-12","2025-03-12")
)
# Convert to Date learning$Login_Date <-
as.Date(learning$Login_Date)
# Extract Month learning$Month <-
format(learning$Login_Date, "%Y-%m")
# Average Quiz Score per Month
monthly_avg <- learning %>%
group_by(Month) %>%
```

```

summarise(Average_Score =
mean(Quiz_Score))
# Moving Average
monthly_avg$Moving_Average <-
stats::filter( monthly_avg$Average_Score,
  rep(1/2, 2),
sides = 1
)
# Line Chart ggplot(monthly_avg, aes(x = Month, y =
Average_Score, group = 1)) + geom_line(color = "blue",
linewidth = 1) + geom_point(size = 3, color = "red") +
geom_line(aes(y = Moving_Average),
  color = "green",
linewidth = 1,      linetype =
"dashed") + labs( title = "Average
Monthly Quiz Score", x =
"Month", y = "Average Quiz
Score"
) +
theme_minimal()

```

Output:



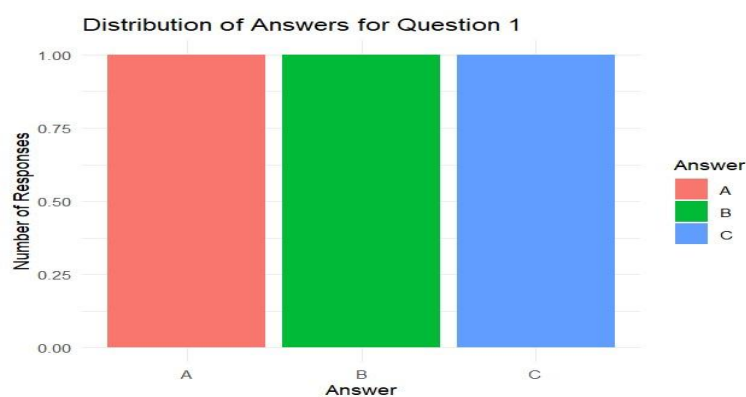
Set 10: Survey Responses Analysis

1. Create a grouped bar chart to visualize the distribution of answers for Question 1. Label the chart elements.

Program:

```
# Load library
library(ggplot2) #
Create dataset
survey <-
data.frame(
  Survey_ID = c(1, 2, 3),
  Question1 = c("A", "B", "C"),
  Question2 = c("B", "A", "A"),
  Question3 = c("C", "D", "B")
)
# Count responses for Question 1 q1 <-
as.data.frame(table(survey$Question1))
colnames(q1) <- c("Answer", "Count")
# Grouped Bar Chart ggplot(q1, aes(x = Answer, y
= Count, fill = Answer)) + geom_bar(stat =
"identity") + labs( title = "Distribution of
Answers for Question 1", x = "Answer", y =
"Number of Responses"
) +
theme_minimal()
```

Output:



2. Generate a stacked bar chart to represent the overall distribution of responses for all three questions.

Program:

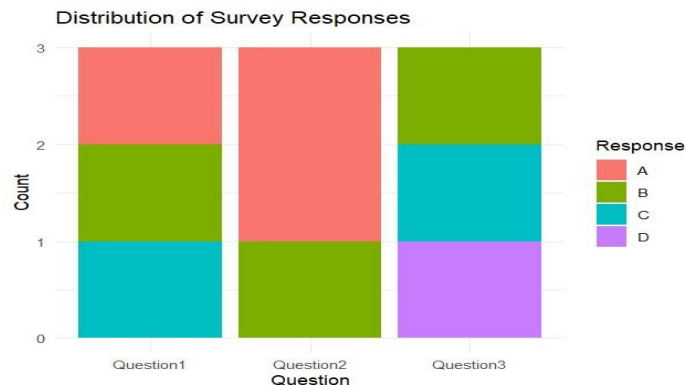
```
# Load libraries

library(ggplot2)
library(tidyr) #
Create dataset
survey <-
data.frame(
  Survey_ID = c(1,2,3),
  Question1 = c("A","B","C"),
  Question2 = c("B","A","A"),
  Question3 = c("C","D","B")
)
# Convert to long format survey_long <-
pivot_longer( survey, cols =
c(Question1, Question2, Question3),
names_to = "Question", values_to =
"Response"
)
# Stacked Bar Chart
ggplot(survey_long,
aes(x = Question,
fill = Response)) +
geom_bar() +

labs( title = "Distribution of Survey
Responses", x = "Question", y =
"Count"
) +
```

theme_minimal()

Output:



3. Build a table to show the survey response data for each survey. Label the table elements.

Program:

```
# Create dataset
```

```
survey <-
```

```
data.frame(
```

```
  Survey_ID = c(1,2,3),
```

```
  Question1 = c("A","B","C"),
```

```
  Question2 = c("B","A","A"),
```

```
  Question3 = c("C","D","B")
```

```
)
```

```
# Display table
```

```
print(survey)
```

Output:

```
Survey_ID Question1 Question2 Question3
```

```
1      1      A      B      C
```

```
2      2      B      A      D
```

```
3      3      C      A      B
```

Set 11: Product Category Analysis

1. Generate a funnel chart to analyze the sales conversion process for each product category. Label the stages and title the chart.

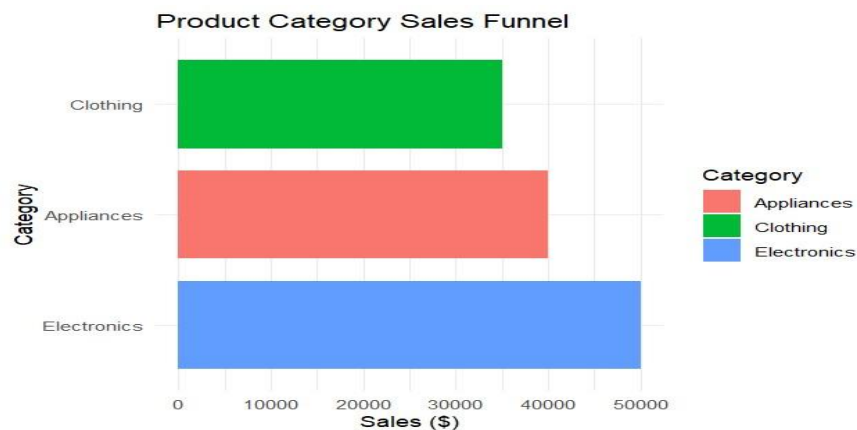
Program:


```

# Load library
library(ggplot2) #
Create dataset
category <-
data.frame(
  Category = c("Electronics","Clothing","Appliances"),
  Sales = c(50000,35000,40000)
)
# Funnel Chart ggplot(category,
aes(x = reorder(Category, -Sales),
y = Sales,      fill = Category)) +
geom_bar(stat = "identity", width =
0.8) + coord_flip() + labs(  title =
"Product Category Sales Funnel",  x
= "Category",  y = "Sales ($)"
) +
theme_minimal()

```

Output:



2. Build a table to display the sales data for each product category. Label the table elements.

Program:

```

# Create dataset
category <-
data.frame(

```

```

Category = c("Electronics","Clothing","Appliances"),
Sales = c(50000,35000,40000)
)

```

```
# Display table
```

```
print(category)
```

Output:

```
Category Sales
```

```
1 Electronics 50000
```

```
2 Clothing 35000
```

```
3 Appliances 40000
```

3. Create a pie chart to represent the distribution of sales across product categories. Include labels.

Program:

```
# Load library
```

```
library(ggplot2) #
```

```
Create dataset
```

```
category <-
```

```
data.frame(
```

```
  Category = c("Electronics", "Clothing", "Appliances"),
```

```
  Sales = c(50000, 35000, 40000)
```

```
)
```

```
# Pie Chart ggplot(category, aes(x = "", y = Sales,
```

```
fill = Category)) + geom_bar(stat = "identity",
```

```
width = 1) + coord_polar(theta = "y") +
```

```
geom_text( aes(label = paste(Category, "\n$",
```

```
Sales)), position = position_stack(vjust = 0.5),
```

```
size = 4
```

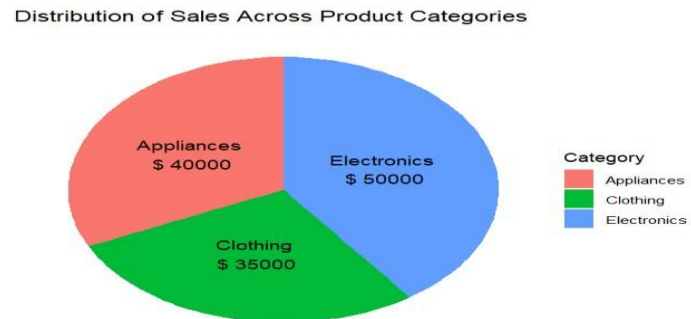
```
) +
```

```
labs( title = "Distribution of Sales Across Product
```

```
Categories"
```

```
) +  
theme_void()
```

Output:



Set 12: Dataset: Website Traffic

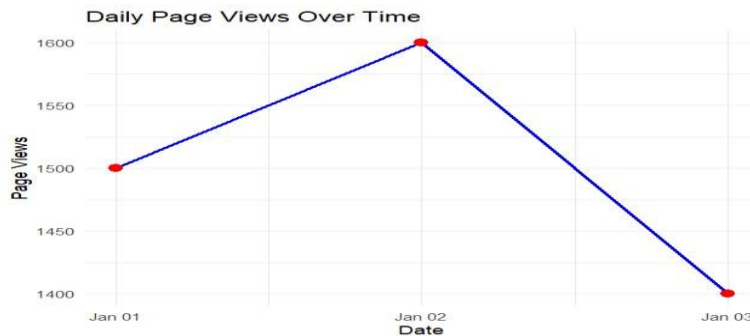
1. Create a line chart to visualize the trend in daily page views over time. Label the axes and title the chart.

Program:

```
# Load library  
  
library(ggplot2) #  
  
Create dataset  
  
traffic <-  
data.frame(  
  Date = as.Date(c("2023-01-01",  
                  "2023-01-02",  
                  "2023-01-03")),  
  Page_Views = c(1500,1600,1400),  
  Click_Through_Rate = c(2.3,2.7,2.0)  
)  
  
# Line Chart ggplot(traffic, aes(x =  
Date, y = Page_Views,  
group = 1)) + geom_line(color =  
"blue", linewidth = 1) +  
geom_point(color = "red", size = 3) +
```

```
labs(title = "Daily Page Views
Over Time", x = "Date", y =
"Page Views"
) +
theme_minimal()
```

Output:



2. Generate a bar chart showing the top N days with the highest click-through rates. Label the chart elements.

Program:

```
# Load library
library(ggplot2) #
Create dataset
traffic <-
data.frame(
  Date = as.Date(c("2023-01-01",
                    "2023-01-02",
                    "2023-01-03")),
  Click_Through_Rate = c(2.3,2.7,2.0)
)
# Sort data top_days <- traffic[order(-
traffic$Click_Through_Rate), ]

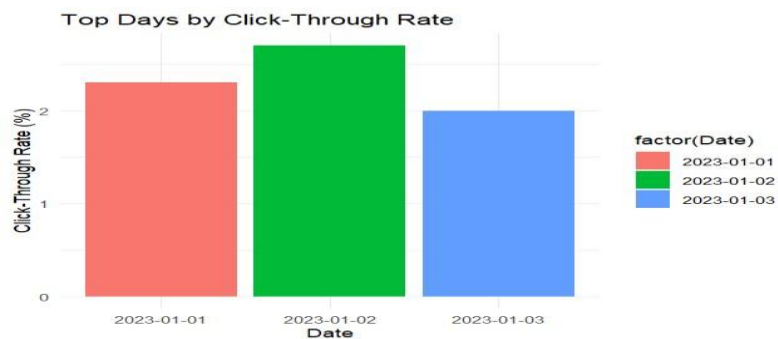
# Bar Chart ggplot(top_days,
aes(x = factor(Date), y
= Click_Through_Rate,
```

```

fill = factor(Date))) +
geom_bar(stat = "identity") +
labs(title = "Top Days by
Click-Through Rate", x =
"Date", y = "Click-
Through Rate (%)")
) +
theme_minimal()

```

Output:



3. Build a table to show the website traffic data for each day. Label the table elements.

Program:

```

# Create dataset
traffic <-
data.frame(
  Date = c("2023-01-01",
           "2023-01-02",
           "2023-01-03"),
  Page_Views = c(1500,1600,1400),
  Click_Through_Rate = c("2.3%","2.7%","2.0%")
)
# Display table
print(traffic)

```

Output:

```

  Date Click_Through_Rate

```

1	2023-01-01	2.3
2	2023-01-02	2.7
3	2023-01-03	2.0